

Software Proposal:

Proposed Software Blueprint

Faculty Workload Allocation, Execution & Effectiveness Management System (FW-EEMS)

1. Purpose of the Software

The system is intended to:

- **Allocate faculty workload** based on institutional policies (40-hour model, differentiated roles)
 - **Ensure execution** of allotted work (not just allocation)
 - **Measure effective utilization of time**
 - **Derive faculty effectiveness metrics** (not simplistic “efficiency”)
 - **Provide evidence for governance, appraisal, and accreditation**
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2. Appropriate Terminology (Very Important)

Instead of using only *efficiency* (which implies speed/output), the following terms are **academically appropriate**:

Recommended Core Metric Name

👉 **Faculty Academic Effectiveness Index (FAEI)**

Supporting Metrics

- **Time Utilization Index (TUI)**
- **Workload Fulfilment Ratio (WFR)**
- **Academic Contribution Score (ACS)**
- **Role Responsibility Fulfilment (RRF)**

These avoid the perception of faculty being treated as factory labor and are more acceptable in academic ecosystems.

3. Major Software Modules

Module 1: Faculty Profile & Role Management

Purpose: Define who the faculty member is and what roles they hold.

Key Data Elements

- Faculty ID, designation, department

- Teaching role (UG/PG/PhD)
- Administrative roles (HoD, Dean, Admissions, CDC, etc.)
- Role tenure (start–end dates)
- Applicable workload model (Standard / Administrative / Full-time Role)

✦ *This module determines which workload distribution table applies.*

Module 2: Workload Policy & Rule Engine

Purpose: Encode institutional workload rules into the system.

Capabilities

- Define workload categories:
 - Teaching
 - AAV
 - Mentoring
 - Administration
 - Research
- Assign percentage & hour limits
- Role-based overrides (Dean, HoD, Admissions Lead)
- Semester-wise or annual policies

✦ *This acts as the brain of the system.*

Module 3: Automated Workload Allocation Engine

Purpose: Convert policy into **individual weekly faculty workload plans**.

Inputs

- Faculty profile
- Timetable constraints
- Academic calendar
- Role responsibilities

Outputs

- Weekly workload breakup (hours & activities)
- Individual faculty timetable (draft)
- Department-level workload summary

✦ *Manual changes allowed only with approval and audit trail.*

Module 4: Execution & Activity Logging Module

Purpose: Ensure **work done = work logged**, without micromanagement.

How it Works

Faculty log activities under predefined categories:

- Teaching sessions (auto-fetched from timetable)
- Assessment / valuation tasks
- Mentoring sessions (student-wise or batch-wise)
- Research activities (paper writing, proposal, experiments)
- Administrative actions (meetings, admissions events, reports)

Features

- Simple daily/weekly logging
- Evidence attachment (optional)
- Mobile & desktop friendly

✦ *This is NOT attendance; it is academic activity logging.*

Module 5: Time Utilization & Fulfilment Analytics

Purpose: Compare **planned vs executed** workload.

Key Calculations

- Planned Hours vs Actual Logged Hours
- Category-wise fulfilment %
- Role-wise deviation analysis
- Peak-load detection (overburdened faculty)

Outputs

- Time Utilization Index (TUI)
 - Workload Fulfilment Ratio (WFR)
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Module 6: Academic Effectiveness & Contribution Engine

Purpose: Translate time + outcomes into **meaningful academic indicators**.

Sample Inputs

- Teaching delivery completion
- Student feedback (normalized)
- Assessment timeliness
- Mentoring outcomes
- Research outputs (not raw counts)
- Administrative deliverables

Key Output

Faculty Academic Effectiveness Index (FAEI)

A weighted composite score, customizable by institution.

✦ *This avoids reducing faculty to “hours only”.*

Module 7: Dashboards & Decision Support

Purpose: Role-based visibility without overexposure.

Dashboards

- Faculty (self-view)
- HoD
- Dean / Director
- Registrar / VC

Insights

- Under-utilization / over-utilization
 - Faculty load imbalance
 - Departmental stress points
 - Evidence for appraisal & promotions
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Module 8: Compliance, Accreditation & Reporting

Purpose: One-click evidence generation.

Reports

- NAAC Criterion-wise faculty engagement
- NBA workload compliance
- Faculty contribution statements
- Departmental workload audit reports

✦ *This module alone saves hundreds of man-hours.*

4. High-Level Software Flow (Textual)

Faculty Profile + Role Assignment



Workload Policy Engine



Automated Workload Allocation



Faculty Weekly Plan & Timetable



Activity Execution & Logging



Time Utilization & Fulfilment Analysis



Academic Effectiveness Index Calculation



Dashboards, Appraisal & Accreditation Reports

5. Software Architecture (Conceptual)

Architecture Type

👉 **Modular, Role-Based, Rule-Driven Web Application**

Suggested Layers

- **Presentation Layer** – Web & Mobile UI
 - **Business Logic Layer** – Workload rules, analytics, scoring
 - **Data Layer** – Faculty, workload, logs, evidence
 - **Integration Layer** – LMS, ERP, HRMS (optional)
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6. Governance & Safeguards (Critical)

- No minute-level tracking (academic dignity preserved)
- Focus on **outcomes + accountability**

- Role-based access control
 - Full audit trail
 - Policy changes versioned
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7. Suggested Software Name Options

- **FAEMS** – Faculty Academic Effectiveness Management System
- **F-WISE** – Faculty Workload & Institutional Support Engine
- **ACADEMETRIX** – Academic Workload & Effectiveness Platform
- **PROFAC** – Professional Faculty Contribution System

3. Software Requirement Specification (SRS)

Faculty Workload Allocation, Execution & Academic Effectiveness Management System (FAEMS)

1. Introduction

1.1 Purpose

The purpose of this software is to **systematically allocate, monitor, and evaluate faculty workload** in alignment with institutional policies, ensuring:

- Fair and transparent workload allotment
- Effective execution of assigned academic and administrative responsibilities
- Measurable utilization of faculty time
- Evidence-based evaluation of faculty academic effectiveness

The system shall support **teaching, research, mentoring, administration, admissions, marketing, student affairs, and career development activities.**

1.2 Scope

The system shall:

- Operate on a **40-hour weekly workload framework**
 - Support **multiple workload models** (regular faculty, HoD, Dean, administrative roles)
 - Capture **planned vs executed work**
 - Compute **academic effectiveness indices**
 - Generate reports for **governance, appraisal, NAAC/NBA accreditation**
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1.3 Definitions (Key Terms)

Term	Definition
FAEI	Faculty Academic Effectiveness Index (composite indicator)
TUI	Time Utilization Index
WFR	Workload Fulfilment Ratio
AAV	Academic preparation, Assessment, Valuation, Documentation

Term	Definition
Role-based workload	Workload distribution based on institutional responsibility

2. User Roles

Role	Access Level
Faculty Member	View plan, log work, self-dashboard
HoD	Department view, approvals, monitoring
Dean / Director	School-level oversight
Registrar / VC	Institutional analytics
Admin (System)	Configuration & policy management

3. Functional Requirements

3.1 Faculty Profile & Role Management

- Create and manage faculty profiles
 - Assign multiple roles (teaching + administrative)
 - Define role tenure (start/end dates)
 - Map faculty to workload model automatically
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3.2 Workload Policy & Rule Engine

- Define workload categories and percentages
 - Support multiple workload templates
 - Allow policy versioning (semester/year-wise)
 - Prevent policy violation (hard constraints)
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3.3 Automated Workload Allocation

- Generate weekly workload plans per faculty
- Allocate hours across categories
- Generate draft faculty timetable

- Allow role-based teaching load reduction
 - Approval workflow (HoD/Dean)
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3.4 Activity Execution & Logging

- Faculty to log activities under predefined categories
 - Teaching auto-validated via timetable
 - Manual logging for research, mentoring, admin
 - Evidence attachment (optional)
 - Weekly submission with lock-in
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3.5 Time Utilization & Fulfilment Analytics

- Compare planned vs executed hours
 - Category-wise fulfilment
 - Identify under/over-utilization
 - Generate TUI and WFR automatically
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3.6 Academic Effectiveness Engine

- Compute **Faculty Academic Effectiveness Index (FAEI)**
 - Weight-based scoring (configurable)
 - Include qualitative + quantitative inputs
 - Avoid raw hour-based ranking
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3.7 Dashboards & Reports

- Faculty self-dashboard
 - Department workload heatmaps
 - School/institution analytics
 - NAAC/NBA-ready reports
 - Export to PDF/Excel
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4. Non-Functional Requirements

4.1 Usability

- Simple UI (academic-friendly)
- Web + mobile responsive
- Minimal daily input (<5 minutes)

4.2 Performance

- Handle institutional-scale data
- Response time <3 seconds

4.3 Security

- Role-based access control
- Audit trail for all changes
- Secure authentication

4.4 Compliance

- Align with NAAC, NBA, NEP 2020
- Preserve academic dignity (no micromanagement)

5. Data Requirements (High-Level)

Core Entities

- Faculty
- Role
- Workload Policy
- Weekly Plan
- Activity Log
- Effectiveness Metrics
- Reports

6. Academic Effectiveness Metrics (Indicative)

Metric	Description
TUI	Actual hours ÷ Planned hours
WFR	Tasks completed ÷ Tasks assigned
ACS	Academic Contribution Score
RRF	Role Responsibility Fulfilment

Metric	Description
FAEI	Weighted composite index

7. System Constraints & Safeguards

- No minute-by-minute tracking
 - No attendance-only logic
 - Outcome-weighted evaluation
 - Transparent faculty access to own data
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8. Future Enhancements (Optional)

- LMS integration
 - HRMS linkage
 - AI-based workload prediction
 - Faculty burnout alerts
 - Research outcome mapping
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9. Deliverables from Vendor / Developer

- Fully functional web application
- Admin configuration panel
- User manuals
- Training & deployment support
- Source code & documentation